

```
> quantile(bank$age, probs = c(0.1, 0.5, 1, 2, 5, 10,
50)/100)
0.1% 0.5% 1% 2% 5% 10% 50%
20.0 22.6 24.0 25.0 27.0 29.0 39.0
```

Interquartile range

The *IQR* function computes the interquartile range for the numeric values of a vector. Its syntax is as follows:

```
IQR(x, na.rm = FALSE, type = 7)
```

The 'type' argument specifies an integer to select the quantile algorithm, which is discussed in Hyndman and Fan (1996). The interquartile range for the bank age can be computed as shown below:

```
> IQR(bank$age, na.rm = FALSE, type=7)
16
```

Standard deviation

The *sd* function is used to calculate the standard deviation for a set of values. It accepts a numeric vector 'x' and a logical value for 'na.rm' to remove missing values. Its syntax is as follows:

```
sd(x, na.rm = FALSE)
```

The standard deviation for a vector with length zero or 1 is 'NA'. A couple of examples are given below:

```
> sd(1:10)
3.02765
```

```
> sd(1)
NA
```

We can calculate the standard deviation for the bank age column as follows:

```
> sd(bank$age)
10.57621
```

You are encouraged to explore more functions available in the stats packages available in R. **END** 

References

- [1] [Moro et al., 2014] S. Moro, P. Cortez and P. Rita; A Data-Driven Approach to Predict the Success of Bank Telemarketing. Decision Support Systems, Elsevier, 62:22-31, June 2014
- [2] Hyndman, R. J. and Fan, Y. (1996); Sample quantiles in statistical packages, American Statistician 50, 361–365, 10.2307/2684934
- [3] The R Stats Package, <https://stat.ethz.ch/R-manual/R-devel/library/stats/html/00Index.html>

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